

Assessment Test III

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

- Which of the following methods can be adopted for the removal of dust particles from air?
 - By using an electrostatic precipitator.
 - By passing air through a strong acidic solution.
 - By passing air through a strong alkaline solution.
 - By passing air through P_2O_5 .
- By which of the following processes can nitrogen be isolated from liquid air?
 - By fractional crystallization
 - By evaporation
 - By dissolving liquid air in water
 - By fractional distillation
- Which of the following gases is evolved when $(NH_4)_2Cr_2O_7$ is heated?
 - N_2
 - O_2
 - NO
 - NO_2
- Assertion (A):** Red phosphorus is less reactive than white phosphorus.
Reason (R): Red phosphorus exists in the form of puckered rings.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true, but R is not the correct explanation of A.
 - A is true and R is false.
 - A is false and R is true.
- Assertion(A):** In the laboratory preparation of chlorine, MnO_2 is used.
Reason(R): MnO_2 is used as a catalyst in the laboratory preparation of chlorine.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true, but R is not the correct explanation of A.
 - A is true and R is false.
 - A is false and R is true.
- Arrange the following forms of sulphur in the ascending order of temperature at which they exist.

(A) μ sulphur	(B) Rhombic sulphur
(C) Monoclinic sulphur	(D) Lambda sulphur
(E) Flower of sulphur	
(a) BCDAE	(b) CDBAE
(c) ACBDE	(d) EDBAC
- Arrange the following processes involved in contact process in a proper sequence.

- (A) Dissolution of an oxide in an acid
 (B) Hydrolysis (reaction with water) of an acid
 (C) Combustion of a substance
 (D) A reversible redox reaction
 (a) DCAB (b) BCAD (c) CDAB (d) ACBD
8. Which of the following elements is not present in bleaching powder?
 (a) Ca (b) O (c) H (d) Cl
9. The coefficient of thermal expansion of which of the following types of glasses is low?
 (a) Borosilicate glass (b) Lead glass
 (c) Optical glass (d) Alkali silicate glass
10. Which of the following fertilizers cannot replenish the soil with nitrogen?
 (a) Ammonium sulphate (b) Urea
 (c) Superphosphate (d) Ammonium phosphate
11. Which of the following processed glasses is used for reinforcing purpose?
 (a) Laminated glass (b) Fibre glass
 (c) Foam glass (d) Opaque glass
12. Which of the following polymers contains cross-linked polymeric chains?
 (a) Polyethylene (b) PVC
 (c) Teflon (d) Melamine
13. Which of the following fibres does not originate from living beings?
 (a) Silk (b) Wool
 (c) Cotton (d) Acrylic
14. Which of the following gases is widely used as an analytical reagent?
 (a) Cl_2 (b) H_2S (c) SO_2 (d) O_2
15. Match the entries of Column A with those of Column B.

Column A (Compounds)	Column B (Uses)
(i) NH_4Cl	(A) Dehydrating agent
(ii) P_2O_5	(B) Flux in the extraction of metals like iron
(iii) CaO	(C) Preparation of soaps and detergents
(iv) CaOCl_2	(D) Electrolyte in dry cell
(v) NaOH	(E) In the manufacture of chloroform

- (a) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (E); (v) $\rightarrow$ (C)
 (b) (i) $\rightarrow$ (D); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (C); (v) $\rightarrow$ (E)
 (c) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (B); (v) $\rightarrow$ (D)
 (d) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (A); (iv) $\rightarrow$ (D); (v) $\rightarrow$ (B)

Assessment Test IV

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. In which of the following ways can CO_2 be removed from air?
 - (a) By using an electrostatic precipitator.
 - (b) By passing air through a strong acidic solution.
 - (c) By passing air through a strong alkaline solution.
 - (d) By passing air through P_2O_5 .
2. Which of the following principles is used to separate nitrogen from liquid air?
 - (a) Difference in freezing points of nitrogen and oxygen
 - (b) Difference in rates of diffusion of nitrogen
 - (c) Difference in solubilities of nitrogen and oxygen in water
 - (d) Difference in the boiling points of liquid nitrogen and liquid oxygen

3. When excess of ammonia gas is made to react with chlorine, _____ and _____ are formed.

- | | |
|--|---|
| (a) NCl_3, H_2 | (b) N_2, HCl |
| (c) $\text{N}_2, \text{NH}_4\text{Cl}$ | (d) $\text{N}_2, \text{H}_2, \text{Cl}_2$ |

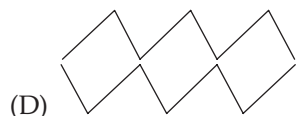
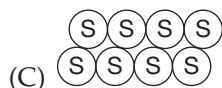
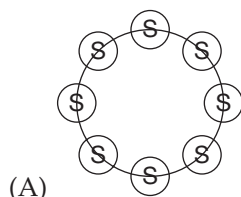
4. **Assertion (A):** Monoclinic sulphur is less dense than its allotrope rhombic sulphur.

Reason (R): The shapes of crystals of monoclinic sulphur and rhombic sulphur are needle-shaped and octahedral, respectively.

- (a) Both A and R are true and R is the correct explanation of A.
 - (b) Both A and R are true, but R is not the correct explanation of A.
 - (c) A is true and R is false.
 - (d) A is false and R is true.
5. **Assertion (A):** HCl gets oxidized to form MnCl_2 during the laboratory preparation of chlorine.

Reason (R): MnO_2 acts as an oxidizing agent in the laboratory preparation of chlorine.

 - (a) Both A and R are true and R is the correct explanation of A.
 - (b) Both A and R are true, but R is not the correct explanation of A.
 - (c) A is true and R is false.
 - (d) A is false and R is true.
 6. Arrange the following forms of sulphur (represented diagrammatically) obtained during the heating of sulphur from room temperature.



(a) DCAB

(b) BDAC

(c) CDBA

(d) ADCB

7. Arrange the following substances used in different steps for the industrial preparation of H_2SO_4 .

(A) Sulphuric acid

(B) O_2 and V_2O_5 (Catalyst)

(C) Water

(D) O_2

(a) CDAB

(b) DBAC

(c) ACBD

(d) BADC

8. Which of the following gases is used in the manufacture of bleaching powder?

(a) Cl_2

(b) O_2

(c) H_2

(d) H_2O_2

9. Which of the following types of glass is soluble in water?

(a) Borosilicate glass

(b) Lead glass

(c) Optical glass

(d) Alkali silicate glass

10. Which of the following fertilizers replenishes the soil with a metal that plays an important role in the growth of plants?

(a) Urea

(b) Ammonium sulphate

(c) Ammonium phosphate

(d) Potassium chloride

11. Which of the following types of processed glass is light in weight and used for civil construction and insulation purposes?

(a) Laminated glass

(b) Fibre glass

(c) Foam glass

(d) Opaque glass

12. A polymer is widely used in lining cooking pans with a non-stick coating. Which of the following statements is not true regarding this polymer?

(a) It is a cross-linked polymer.

(b) It is made up of linearly linked polymeric units.

(c) It is resistant to chemicals.

(d) It is heat-resistant.

13. Which of the following synthetic fibres are similar to the natural fibres, silk, and wool, respectively?

(a) Nylon and rayon

(b) Acrylic and rayon

(c) Polyester and acrylic

(d) Rayon and acrylic

14. Which of the following gases is a reducing agent and is used to identify certain metals present in the given salt?

- (a) Cl_2 (b) H_2S (c) SO_2 (d) O_2

15. Match the entries of Column A with those of Column B.

Column A	Column B
(i) Electrolysis of saturated brine solution	(A) CaOCl_2
(ii) Thermal decomposition of limestone	(B) NH_4Cl
(iii) On decomposition produces two gases	(C) CaO
(iv) Produces nascent oxygen	(D) Conc. H_2SO_4
(v) Removes hydrogen and oxygen in the form of a water molecule from a compound	(E) NaOH

- (a) (i) $\rightarrow$ (E); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (A); (v) $\rightarrow$ (D)
 (b) (i) $\rightarrow$ (E); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (D); (v) $\rightarrow$ (A)
 (c) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (A); (v) $\rightarrow$ (E)
 (d) (i) $\rightarrow$ (B); (ii) $\rightarrow$ (C); (iii) $\rightarrow$ (D); (iv) $\rightarrow$ (E); (v) $\rightarrow$ (A)

Space for rough work

Answer Keys

Assessment Test I

1. (c) 2. (a) 3. (a) 4. (c) 5. (a) 6. (a) 7. (c) 8. (b) 9. (c) 10. (c)
 11. (a) 12. (a) 13. (a) 14. (a) 15. (b)

Assessment Test II

1. (b) 2. (a) 3. (a) 4. (a) 5. (c) 6. (b) 7. (b) 8. (c) 9. (b) 10. (b)
 11. (c) 12. (b) 13. (c) 14. (c) 15. (c)

Assessment Test III

1. (a) 2. (d) 3. (a) 4. (c) 5. (c) 6. (a) 7. (c) 8. (c) 9. (a) 10. (c)
 11. (b) 12. (d) 13. (d) 14. (b) 15. (a)

Assessment Test IV

1. (c) 2. (d) 3. (c) 4. (b) 5. (d) 6. (b) 7. (b) 8. (a) 9. (d) 10. (d)
 11. (c) 12. (a) 13. (d) 14. (b) 15. (a)

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Hints and Explanations

CHAPTER 1 Atomic Structure

Assessment Test I

1. Discovery of isotopes led to contradiction of the postulate of Dalton's atomic theory that atoms of the same element are alike in all aspects.

Hence, the correct option is (a).

2. The figure represents Thomson's atomic model. From Thomson's experiment, it can be concluded that electrons are the universal constituent of matter.

Hence, the correct option is (d).

3. The energy of orbits increases with distance from the nucleus in the order $K < L < M < N$. Energy is released when an electron jumps from a higher energy orbit to a lower energy orbit. Among the given transitions, such transitions from higher orbit to lower orbit are $L \rightarrow K$, $M \rightarrow L$, and $N \rightarrow K$.

Hence, the correct option is (c).

4. The orbits in which electrons revolve around the nucleus are called stationary orbits since the energy of electrons remains constant in those orbits. The electrons are in continuous motion around the nucleus in those orbits and never remain stationary.

Hence, the correct option is (c).

5. Mass of electron = m

Mass of proton = $1837 \times m$

No. of protons = $\frac{36,740}{1837} = 20$

No. of neutrons = 19

Mass number = 39

Atomic number = 20

Representation of the atom or nuclide $\rightarrow {}_{20}\text{X}^{39}$.

Hence, the correct option is (d).

6. No. of electrons in a dispositive ion = a

No. of electrons in neutral atom = $a + 2$

Atomic number = No. of protons = $a + 2 = b$

No. of neutrons = $a + 2 + 3 = a + 5$

Mass number = $(a + 2) + (a + 5) = 2a + 7$

Hence, the correct option is (b).

7. O^{-2} ion is formed by the gain of two electrons by neutral atom and K^{+} ion is formed by the loss of one electron by neutral atom.

Hence, the correct option is (c).

8. (i) The sum of the number of protons and neutrons is called mass number.

(ii) Rutherford's theory proposed the concept of nucleus.

(iii) Isotopes differ in the number of neutrons.

(iv) Isobars possess same number of nucleons.

Hence, the correct option is (b).

9. A few of the α particles in the experiment were completely rebounded because they hit the nucleus that is a positively small central part and experience repulsion from the positively the α particles charged nucleus.

Hence, the correct option is (d).

10. (i) Combination of atoms in a simple numerical ratio.
 (ii) Electrical neutrality of atom.
 (iii) The diameter of nucleus as 10^{-5} times that of an atom.
 (iv) Presence of electrons in K, L, M, and N shells.
 Hence, the correct option is (c).

11. (E) ${}_{12}\text{X}^{24}$ & ${}_{20}\text{Y}^{40} \rightarrow$ Metals
 (D) ${}_8\text{X}^{16}$ & ${}_{16}\text{Y}^{32} \rightarrow$ Non-metals
 (B) ${}_{10}\text{X}^{20}$ & ${}_{18}\text{Y}^{36} \rightarrow$ Noble gases
 (C) ${}_Z\text{A}^X$ & ${}_Z\text{A}^{X+2} \rightarrow$ Isotopes
 (A) ${}_Z\text{A}^X$ & ${}_{Z+2}\text{B}^X \rightarrow$ Isobars

Hence, the correct option is (a).

12. (i) $\rightarrow$ (C)
 $Z = 21$ corresponds to electronic configuration 2, 8, 9, and 2. The ratio of electrons in penultimate shell to valence shell is 9:2.
 (ii) $\rightarrow$ (E)
 $Z = 30$ corresponds to electronic configuration 2, 8, 18, and 2. The ratio of electrons in penultimate shell to valence shell is 9:1.
 (iii) $\rightarrow$ (B)
 $Z = 20$ corresponds to electronic configuration 2, 8, 8, and 2. The ratio of electrons in penultimate shell to valence shell is 4:1.
 (iv) $\rightarrow$ (A)
 $Z = 19$ corresponds to electronic configuration 2, 8, 8 and 1. The ratio of electrons in penultimate shell to valence shell is 8:1.
 (V) $\rightarrow$ (D)
 $Z = 18$ corresponds to electronic configuration 2, 8 and 8. The ratio of electrons in penultimate shell to valence shell is 1:1.

Hence, the correct option is (a).

13. The anion formed from the given neutral atom is a dinegative ion which has the configuration of argon the electronic configuration of the element is 2, 8 and 6.

Hence, the correct option is (c).

14. The electrons number 36 does not correspond to $2n^2$ which gives the maximum number of electrons in Bohr's orbit.

Hence, the correct option is (b).

15. Y can have an atomic number 30. The electronic configuration for $Z = 30$ is 2, 8, 18, and 2. On losing two electrons, the resultant dipositive ion gets the configuration of 2, 8, and 18, and does not have an octet. X is K(19) and its electronic configuration is 2, 8, 8, and 1. On losing one electron, the resultant mono positive ion gets the configuration of 2, 8, and 8.

Hence, the correct option is (a).

Assessment Test II

1. Law of definite proportions provided experimental basis for the postulate of Dalton's atomic theory that atoms of different elements combine in a simple whole number ratio.

Hence, the correct option is (c).

2. The given atomic model is Bohr's atomic model, which supported the presence of a nucleus at the centre of an atom. Thus, it contradicted the idea of uniform distribution of positive charge in an atom.

Hence, the correct option is (d).

3. In the given figure, the naming of orbits is in the order M, N, L, and K, whereas the usual order is K, L, M, and N, which corresponds to 1, 2, 3, and 4. All the other aspects are in accordance with the established facts.

Hence, the correct option is (c).

4. The energy of orbits increases with increase in distance because the nuclear force of attraction decreases with increase in distance. Since the first orbit is associated with the lowest energy, electrons in that orbit do not lose energy at all. Both the statements are correct.

Hence, the correct option is (b).

5. No. of protons = $\frac{18370}{1837} = 10$

Atomic number = 10

Mass number = $20 + 2 = 22$

No. of neutrons = $22 - 10 = 12$.

Hence, the correct option is (d).

6. Mass of electron = x

Mass of protons = y

No. of protons in ${}_8\text{O}^{18} = 8$

Mass of one proton = $y/8 = 1837x$

$$y = 8 \times 1837x$$

Hence, the correct option is (c).

7. Neutral atom with 7 electrons in valence shell corresponds to the formation of uninegative ion as it can gain one electron to get an octet. A neutral atom with 3 electrons in valence shell corresponds to the formation of tripositive ion as it can lose 3 electrons to get an octet.

Hence, the correct option is (b).

8. (i) The number of neutrons in ${}_{17}\text{Cl}^{35}$ is 18.
 (ii) Maximum number of electrons in an orbit can be given by $2n^2$.
 (iii) ${}_{19}\text{Y}^{40}$ is the isobar of ${}_{20}\text{X}^{40}$.
 (iv) Isotope of ${}_Z\text{X}^A$ is ${}_Z\text{Y}^{A+2}$.

Hence, the correct option is (c).

9. Both the statements are true. Most of the α particles in the experiment pass straight through the gold foil which signifies that most part of the atom is empty.

Hence, the correct option is (a).

10. (i) Discovery of electrons
 (ii) Discovery of nucleus
 (iii) Establishing stability of atom
 (iv) Discovery of protons

Hence, the correct option is (b).

11. (B) $B \rightarrow 2, 8; Z = 10 \rightarrow$ Noble gas
 (A) $A \rightarrow 2, 8; Z = 11 \rightarrow$ Cation
 (E) $E \rightarrow 2, 8, 5; Z = 15 \rightarrow$ Non-metal atom
 (C) $C \rightarrow 2, 8, 8; Z = 16 \rightarrow$ Anion
 (D) $D \rightarrow 2, 8, 2; Z = 12 \rightarrow$ Metal atom

Hence, the correct option is (d).

12. (i) $\rightarrow$ (B)

$Z = 36$ corresponds to electronic configuration 2, 8, 18, and 8. There are 18 electrons in the penultimate shell.

- (ii) $\rightarrow$ (D)

$Z = 17$ corresponds to electronic configuration 2, 8, and 7. There are 8 electrons in the penultimate shell.

- (iii) $\rightarrow$ (E)

$Z = 25$ corresponds to electronic configuration 2, 8, 13, and 2. There are 13 electrons in the penultimate shell.

- (iv) $\rightarrow$ (A)

$Z = 9$ corresponds to electronic configuration 2 and 7. There are 2 electrons in the penultimate shell.

- (v) $\rightarrow$ (C)

$Z = 22$ corresponds to electronic configuration 2, 8, 10, and 2. There are 10 electrons in penultimate shell.

Hence, the correct option is (b).

13. The given species is a neutral atom with 13 electrons that means 3 electrons in the valence shell. It can lose three electrons to form a tripositive ion with the configuration of neon.

Hence, the correct option is (a).

14. The maximum number of electrons in an orbit is given by $2n^2$. If $2n^2 = 50$, then $n = 5$.

Hence, the correct option is (c).

15. The electronic configuration of the element is 2, 8, 14, and 2. The atomic number of the element is 26.

Hence, the correct option is (a).

Assessment Test III

1. (A) 82 neutrons
 (B) 146 neutrons
 (C) 118 neutrons
 (D) 142 neutrons
 (E) 117 neutrons
 AECDB

Hence, the correct option is (c).

2. (A) $A^+ \rightarrow 11$ protons $Z = 11$
 (B) $B^{+2} \rightarrow 12$ protons $Z = 12$
 (C) $C^- \rightarrow 9$ protons $Z = 9$
 (D) $D^{-2} \rightarrow 8$ protons $Z = 8$
 DCAB

Hence, the correct option is (a).

3. In the antepenultimate shell of Al, only 2 electrons are present.
Hence, the correct option is (d).
4. Isotopes of an element do not possess the same number of fundamental particles. They possess the same number of protons and electrons but differ with respect to the number of neutrons. Atomic number is the number of protons which is the characteristic of an element.
Hence, the correct option is (d).
5. No. of protons = $\frac{23881}{1837} = 13$
Atomic number = 13.
Hence, the correct option is (c).
6. α particle has 2 protons and 2 neutrons. A proton and a neutron have almost equal mass. So, the mass of α particle is $4m$.
Hence, the correct option is (c).
7. Thomson's model failed because it could not explain how the positive charge and electrons coexist in an atom without getting neutralized.
Hence, the correct option is (c).
8. Number of neutron = Number of proton = Number of electron.
= Atomic number = 20
Mass Number is = $2 \times \text{Atomic number}$
= $20 \times 20 = 40$
So, ${}_{20}\text{X}^{40} \Rightarrow$. This element is Ca.
Hence, the correct option is (a).
9. An atom with $Z = 13$ has the electronic configuration 2, 8, and 3. It can form tripositive ion.
Hence, the correct option is (d).
10. An atom with electronic configuration 2, 8, 18, and 8 is an inert gas and has one incomplete shell. It cannot form ions as it has an octet configuration and does not take part in chemical reactions. The next electron extends into the fifth shell as valence shell can have a maximum of 8 electrons.
Hence, the correct option is (c).

11. $a = 2$ electrons, $a + b = 8$ electrons, $b = 6$ electrons, and $2b = 12$ electrons
The electrons configuration of atom A should be 2, 8, 12, and 2. This element has 2 incomplete shells. Electrons can be added to penultimate shell till 18 electrons are filled in it. An element with $Z + 7$ has 3 electrons in its valence shell. It can form a dipositive ion.
Hence, the correct option is (c).
12. As the electron goes away from the nucleus, the nuclear force of attraction decreases and hence, the electron revolves round with lesser velocity.
Hence, the correct option is (b).
13. The postulate that electrons are in continuous motion around the nucleus was not based on the experimental results.
Hence, the correct option is (c).
14. ${}_{14}\text{A}^{28} \rightarrow 2, 8, 4$
 ${}_6\text{B}^{12} \rightarrow 2, 4$
Hence, the correct option is (b).
15. (i) $\rightarrow$ (C); (ii) $\rightarrow$ (A); (iii) $\rightarrow$ (B); (iv) $\rightarrow$ (D)
Hence, the correct option is (b).

Assessment Test IV

1. (A) $A \rightarrow 16; 2, 8, 6$
(B) $B \rightarrow 20; 2, 8, 8, 2$
(C) $C \rightarrow 14; 2, 8, 4$
(D) $D \rightarrow 17; 2, 8, 7$
(E) $E \rightarrow 13; 2, 8, 3$
DACEB
Hence, the correct option is (d).
2. (A) $A = 87$
(B) $A = 137$
(C) $A = 190$
(D) $A = 108$
(E) $A = 101$
AEDBC
Hence, the correct option is (d).
3. Isotopes have the same atomic number but different mass number.
Hence, the correct option is (c).